

How exactly does AI use up water?

Posted by u/8Bit_Cat • 0 points • 8 comments

Like unless they turn it into hydrogen and oxygen through electrolysis or send it into space where would it be going? How exactly would it get used up? Water sent down the drain gets treated then pumped into the ocean back in the water cycle.

Comments

u/navelencounters • 1 points • 2026-02-21 19:36 UTC

I work with data centers (design and build)...the water is for cooling....it does not use up the water but will need to be treated once it passes through all the pipes....if the center does not have a well, pond or lake to pull from, then they will use municipal water (in large tanks) which costs consumers more.

u/Primary-Basket3416 • 0 points • 2026-02-21 17:47 UTC

AI is stored in computers All electronics love the cold, hate the hot. Ever notice why some reporters and weather people in studios have on some heavy clothes. Their equipment needs AC or it overheats. So data centers need AC to keep cool and running. And what is AC..a water based coolant.

u/Lumpy-Notice8945 • 2 points • 2026-02-21 17:16 UTC

I recomend you watch the youtube video Hank green made about this.

Its a lot of misleading news headlines out there and many are just plain wrong or just miss the point.

Datacenters(and thats by no means exclusive to AI, any website or service runs in datacenters) need some form of cooling, most have some kind of AC installed and some of these ACs use evaporative cooling, so they need tap water and thats turned into steam.

In no case water is physically destroyed or chemically broken down, thats not even the case for regular water use in other cases but it still wont be available as drinkable water and there still need to be water treatment plants that produce clean water from rivers or clean sewage water.

u/explosive-diorama • 1 points • 2026-02-21 17:14 UTC

Water is not permanently destroyed, but clean potable water is used to cool the systems, making it unclean, where it needs to be re-treated at the water plant.

u/8Bit_Cat • 1 points • 2026-02-21 17:15 UTC

Wouldn't a water cooling system be a closed system?

u/illogictc • 2 points • 2026-02-21 17:30 UTC

Not if they utilize evaporative cooling which is a common, and open-loop, method. Because AI centers are gonna be even more demanding in regards to cooling, companies are looking into options to instead be closed-loop. Might have to cycle fresh water in now and again but it wouldn't be a constant use like evaporative cooling is.

u/explosive-diorama • 3 points • 2026-02-21 17:16 UTC

Many times, they are. Not all of them are.

The whole "AI uses water" argument is probably the worst argument out of many good ones, but I guess it scares people? It's like saying getting shot in the face is bad because lead is toxic.

Sure, but also, getting shot in the face is worse than the lead part.

u/Fearlessleader85 • 1 points • 2026-02-21 19:33 UTC

It's not a bad argument, it's just water is not well understood as a resource. Water scarcity is a localized issue. They're are places that have far more water than they know what to do with, and there's places that struggle to keep enough clean drinking water available.

Take a quick google trip to Prineville, Oregon. This is a small town in the Oregon High Desert. It barely has enough water to graze cattle. They've had a bunch of extremely large data centers installed over the last 15 years or so. Their water consumption is a problem for that area. It's not a problem for Charleston, SC, or Denver CO, or even Bend, Oregon just a short drive away. But it is a problem for Prineville and the surrounding area.

In the town that i live, we don't have much problem with water availability, but if you just go 10 miles away, the aquifers are literally collapsing. Not from AI use, but just overuse from irrigation and domestic water while more and more areas that used to be permeable farmland are paved and water that would have filled the aquifers is now running through stormwater treatment and out into the river.